

```
Processando config 24, seed 1, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
```

```
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 0.23
p-value of test : 0.89
```

```
-----
CPU time used : 00:00:00.03
```

```
Generator state:
```

```
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 7.60  
p-value of test : 0.47

-----  
CPU time used : 00:00:00.02

Generator state:

33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 177.87  
p-value of test : 2.4e-3

-----  
CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32543.14  
p-value of test : 0.81

-----  
CPU time used : 00:00:00.05

Generator state:

67108864 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 169.10  
p-value of test : 0.65

-----  
CPU time used : 00:00:00.02

Generator state:

83886080 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 283.93  
p-value of test : 0.80  
-----

CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.67  
p-value of test : 0.25  
-----

CPU time used : 00:00:00.02

Generator state:

117440512 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 30.28 |
| p-value of test              | : | 0.55  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 26.50 |
| p-value of test              | : | 0.70  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 18.56 |
| p-value of test              | : | 0.97  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 16.89 |
| p-value of test              | : | 0.89  |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 15     |
| ChiSquare statistic          | : | 35.83  |
| p-value of test              | : | 1.9e-3 |

-----  
CPU time used : 00:00:00.06

Generator state:

134217728 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 70.47 |
| p-value of test              | : | 0.22  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 62.49 |
| p-value of test              | : | 0.46  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 152.10 |
| p-value of test              | : | 0.66   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 55.37 |
| p-value of test              | : | 0.54  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 36.85 |
| p-value of test              | : | 0.25  |

-----  
CPU time used : 00:00:00.06

Generator state:  
150994944 bits have been read.

===== Summary results of Alphabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.33                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 2.02
p-value of test                    : 0.36
```

```
-----
CPU time used                      : 00:00:00.03
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 8     |
| Value of the statistic       | : | 13.74 |
| p-value of test              | : | 0.09  |

-----  
CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 128    |
| Value of the statistic       | : | 158.62 |
| p-value of test              | : | 0.03   |

-----  
CPU time used : 00:00:00.02

Generator state:

201326592 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 32942.67
p-value of test : 0.25
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

218103808 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring\_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 157.22
p-value of test : 0.85
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:

234881024 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 304.41 |
| p-value of test              | : | 0.50   |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.02 |
|---------------|---|-------------|

Generator state:

251658240 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |      |
|------------------|---|------|
| Normal statistic | : | 0.95 |
| p-value of test  | : | 0.17 |

-----

|               |   |             |
|---------------|---|-------------|
| CPU time used | : | 00:00:00.00 |
|---------------|---|-------------|

Generator state:

268435456 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 48.56 |
| p-value of test              | : | 0.03  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 45.88 |
| p-value of test              | : | 0.04  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 24.19 |
| p-value of test              | : | 0.84  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 17.12 |
| p-value of test              | : | 0.88  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 13.99 |
| p-value of test              | : | 0.53  |

-----  
CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 56.49 |
| p-value of test              | : | 0.67  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 62.64 |
| p-value of test              | : | 0.45  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 164.00 |
| p-value of test              | : | 0.40   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 56.70 |
| p-value of test              | : | 0.49  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 31.10 |
| p-value of test              | : | 0.51  |

-----  
CPU time used : 00:00:00.06

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

|                       |                                                |
|-----------------------|------------------------------------------------|
| Version:              | TestU01 1.2.3                                  |
| Generator:            | ufire_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits:       | 16777216                                       |
| Number of statistics: | 17                                             |
| Total CPU time:       | 00:00:00.28                                    |

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.69
p-value of test                   : 0.71
```

```
-----
CPU time used                     : 00:00:00.02
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin\_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 9.47  
p-value of test : 0.30

-----  
CPU time used : 00:00:00.02

Generator state:  
335544320 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For  $\Delta > -1$ , we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 142.71  
p-value of test : 0.18

-----  
CPU time used : 00:00:00.00

Generator state:  
352321536 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32552.16  
p-value of test : 0.80

-----  
CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 180.22  
p-value of test : 0.42

-----  
CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 309.81 |
| p-value of test              | : | 0.41   |

-----

CPU time used : 00:00:00.02

Generator state:

402653184 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |   |       |
|------------------|---|-------|
| Normal statistic | : | -0.64 |
| p-value of test  | : | 0.74  |

-----

CPU time used : 00:00:00.02

Generator state:

419430400 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 45.52 |
| p-value of test              | : | 0.06  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 28.26 |
| p-value of test              | : | 0.61  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 30.27 |
| p-value of test              | : | 0.55  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 33.76 |
| p-value of test              | : | 0.11  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 10.54 |
| p-value of test              | : | 0.78  |

-----  
CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 74.61 |
| p-value of test              | : | 0.13  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 55.87 |
| p-value of test              | : | 0.69  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 141.33 |
| p-value of test              | : | 0.85   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 68.02 |
| p-value of test              | : | 0.15  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 19.78 |
| p-value of test              | : | 0.96  |

-----  
CPU time used : 00:00:00.06

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

|                 |                                                |
|-----------------|------------------------------------------------|
| Version:        | TestU01 1.2.3                                  |
| Generator:      | ufile_CreateReadBin, unif01_CreateBitBlockGen: |
| Number of bits: | 16777216                                       |

Number of statistics: 17  
Total CPU time: 00:00:00.27

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alhabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 0.24  
p-value of test : 0.89

-----  
CPU time used : 00:00:00.02

Generator state:  
469761792 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 5.62
p-value of test              : 0.69
```

-----  
CPU time used : 00:00:00.00

Generator state:

486539008 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

-----  
Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 133.52
p-value of test              : 0.35
```

-----  
CPU time used : 00:00:00.02

Generator state:

503316224 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32199.86  
p-value of test : 0.99

-----

CPU time used : 00:00:00.02

Generator state:

520093440 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 167.92  
p-value of test : 0.68

-----

CPU time used : 00:00:00.03

Generator state:  
536870656 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 305    |
| Chi-square statistic         | : | 312.57 |
| p-value of test              | : | 0.37   |

-----  
CPU time used : 00:00:00.02

Generator state:  
553647872 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : -0.42  
p-value of test : 0.66

-----  
CPU time used : 00:00:00.00

Generator state:  
570425088 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 22.62 |
| p-value of test              | : | 0.89  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 27.54 |
| p-value of test              | : | 0.64  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 34.56 |
| p-value of test              | : | 0.35  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 23.92 |
| p-value of test              | : | 0.52  |

-----  
Test on the values of the Statistic C

|                              |   |      |
|------------------------------|---|------|
| Number of degrees of freedom | : | 15   |
| ChiSquare statistic          | : | 6.43 |
| p-value of test              | : | 0.97 |

-----  
CPU time used : 00:00:00.06

Generator state:

587202304 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config24_seed1.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 75.71 |
| p-value of test              | : | 0.11  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 63.89 |
| p-value of test              | : | 0.41  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 148.70 |
| p-value of test              | : | 0.73   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 54.17 |
| p-value of test              | : | 0.58  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 18.20 |
| p-value of test              | : | 0.98  |

-----  
CPU time used : 00:00:00.05

Generator state:  
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 0.17  
p-value of test : 0.92

-----  
CPU time used : 00:00:00.02

Generator state:  
620756480 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 14.65  
p-value of test : 0.07

-----  
CPU time used : 00:00:00.02

Generator state:

637533696 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin

unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 120.13  
p-value of test : 0.68

-----  
CPU time used : 00:00:00.02

Generator state:

654310912 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32892.34  
p-value of test : 0.31

-----

CPU time used : 00:00:00.02

Generator state:

671088128 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 179.88  
p-value of test : 0.43

-----

CPU time used : 00:00:00.02

Generator state:  
687865344 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----

|                              |          |
|------------------------------|----------|
| Number of degrees of freedom | : 305    |
| Chi-square statistic         | : 270.06 |
| p-value of test              | : 0.93   |

-----

CPU time used : 00:00:00.00

Generator state:  
704642560 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----

|                  |          |
|------------------|----------|
| Normal statistic | : -0.083 |
| p-value of test  | : 0.53   |

-----

CPU time used : 00:00:00.02

Generator state:  
721419776 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 25.70 |
| p-value of test              | : | 0.78  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 28.10 |
| p-value of test              | : | 0.62  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 24.82 |
| p-value of test              | : | 0.81  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 37.03 |
| p-value of test              | : | 0.06  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 15.48 |
| p-value of test              | : | 0.42  |

-----  
CPU time used : 00:00:00.06

Generator state:

738196992 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 69.58 |
| p-value of test              | : | 0.24  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 66.40 |
| p-value of test              | : | 0.33  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 181.12 |
| p-value of test              | : | 0.12   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 43.56 |
| p-value of test              | : | 0.90  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 29.21 |
| p-value of test              | : | 0.61  |

-----  
CPU time used : 00:00:00.05

Generator state:  
754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3  
Generator: ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits: 16777216  
Number of statistics: 17  
Total CPU time: 00:00:00.25

All tests were passed

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Alphabit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 2,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 4  
Expected number per cell = 4194304  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 2  
Value of the statistic : 2.14  
p-value of test : 0.34

-----

CPU time used : 00:00:00.02

Generator state:

771751168 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 4,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 16  
Expected number per cell = 1048576  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 8  
Value of the statistic : 8.12  
p-value of test : 0.42

-----  
CPU time used : 00:00:00.00

Generator state:

788528384 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----  
N = 1, n = 16777216, r = 0, s = 32, L = 8,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 256  
Expected number per cell = 65536  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

-----  
Test Results for Delta = 1.0000

Number of degrees of freedom : 128  
Value of the statistic : 148.58  
p-value of test : 0.10

-----  
CPU time used : 00:00:00.02

Generator state:  
805305600 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin\_MultinomialBitsOver test:

-----

N = 1, n = 16777216, r = 0, s = 32, L = 16,  
Sparse = FALSE

Number of bits = n = 16777216  
Number of cells =  $2^L$  = 65536  
Expected number per cell = 256  
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation  
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

-----

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768  
Value of the statistic : 32781.27  
p-value of test : 0.48

-----

CPU time used : 00:00:00.03

Generator state:  
822082816 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 209.63  
p-value of test : 0.05

-----  
CPU time used : 00:00:00.02

Generator state:  
838860032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingIndep test:

-----

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 339.55  
p-value of test : 0.08

-----  
CPU time used : 00:00:00.02

Generator state:  
855637248 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring\_HammingCorr test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.40  
p-value of test : 0.35

-----  
CPU time used : 00:00:00.00

Generator state:  
872414464 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 27.85 |
| p-value of test              | : | 0.68  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 31    |
| ChiSquare statistic          | : | 28.58 |
| p-value of test              | : | 0.59  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 21.45 |
| p-value of test              | : | 0.92  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 25    |
| ChiSquare statistic          | : | 15.50 |
| p-value of test              | : | 0.93  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 15    |
| ChiSquare statistic          | : | 12.33 |
| p-value of test              | : | 0.65  |

-----  
CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config24\_seed1.bin  
unif01\_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk\_RandomWalk1 test:

-----  
N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 40.48 |
| p-value of test              | : | 0.98  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 62    |
| ChiSquare statistic          | : | 50.31 |
| p-value of test              | : | 0.86  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 160    |
| ChiSquare statistic          | : | 137.52 |
| p-value of test              | : | 0.90   |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 57    |
| ChiSquare statistic          | : | 67.87 |
| p-value of test              | : | 0.15  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 32    |
| ChiSquare statistic          | : | 21.02 |
| p-value of test              | : | 0.93  |

-----  
CPU time used : 00:00:00.05

Generator state:

905968640 bits have been read.

===== Summary results of Alphabit =====

Version:            TestU01 1.2.3  
Generator:          ufile\_CreateReadBin, unif01\_CreateBitBlockGen:  
Number of bits:     16777216  
Number of statistics: 17  
Total CPU time:     00:00:00.25

All tests were passed